

AIB-2401

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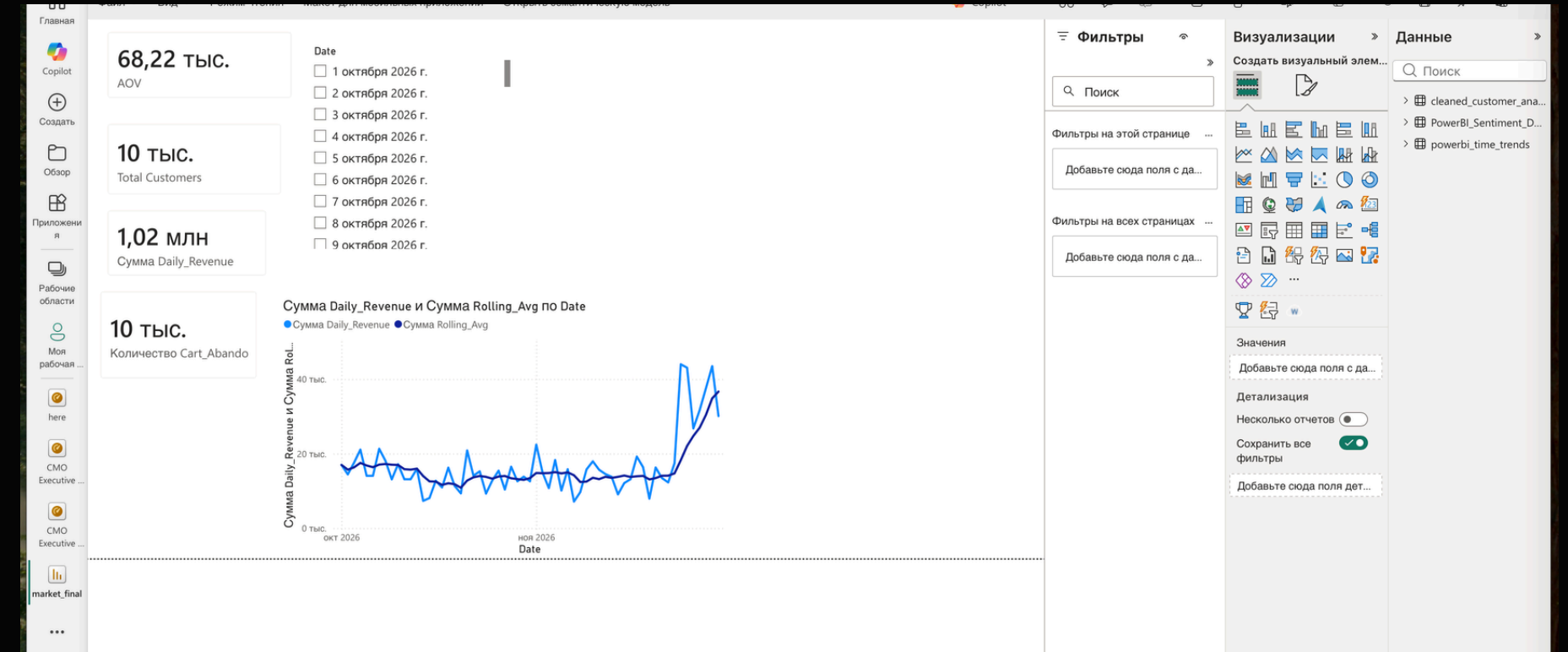
AI-MARKETING

Final project

EXECUTIVE MARKETING OVERVIEW

As seen on Page 1 of our report, this is the executive command center. Here, the CMO can instantly track core KPIs: Total Revenue—derived from the sum of our Monetary data—Average Order Value (Avg_Order_Value), and unique customer counts. The central visualization highlights our revenue trend over time, utilizing a Rolling Average line to smooth out short-term fluctuations and reveal true macro-trends.

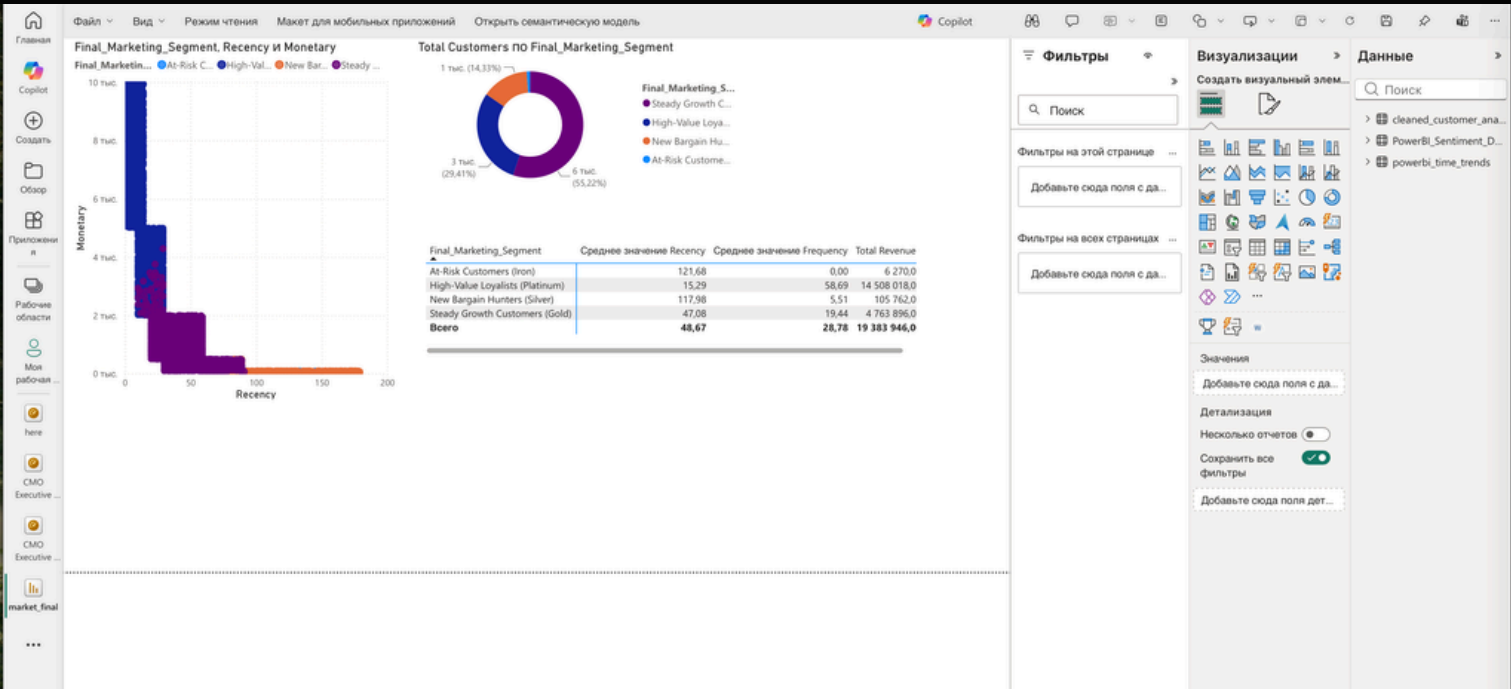
To make this page highly interactive, we integrated a native AI Q&A (Questions & Answers) feature. By leverage natural language processing, the executive can type any business query in plain English—such as show total revenue—and the system automatically renders the accurate financial figure instantly."



CUSTOMER SEGMENTATION DASHBOARD

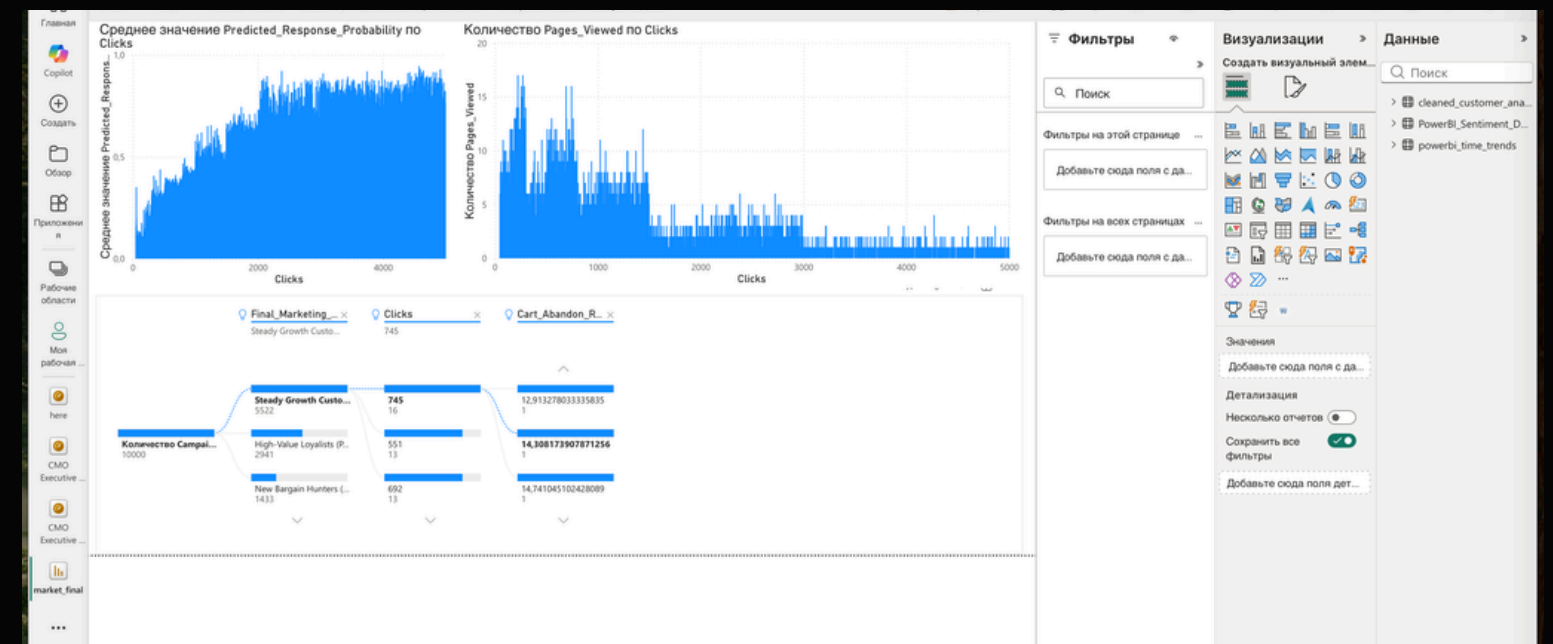
"Moving to Page 2, we present our customer segmentation strategy. Rather than manually grouping customers based on guesswork, we deployed an unsupervised machine learning algorithm: K-Means Clustering. The model evaluated three critical customer behavior vectors simultaneously: Recency (days since last purchase), Frequency (total purchase count), and Monetary (total spend).

The algorithm automatically partitioned our audience into distinct behavioral clusters, which we mapped onto this scatter plot. Furthermore, we configured Advanced Tooltips: hovering over any segment dynamically pulls a custom micro-card displaying specific performance metrics, like the average order value and cart abandonment rate, keeping the main interface clean and scannable."

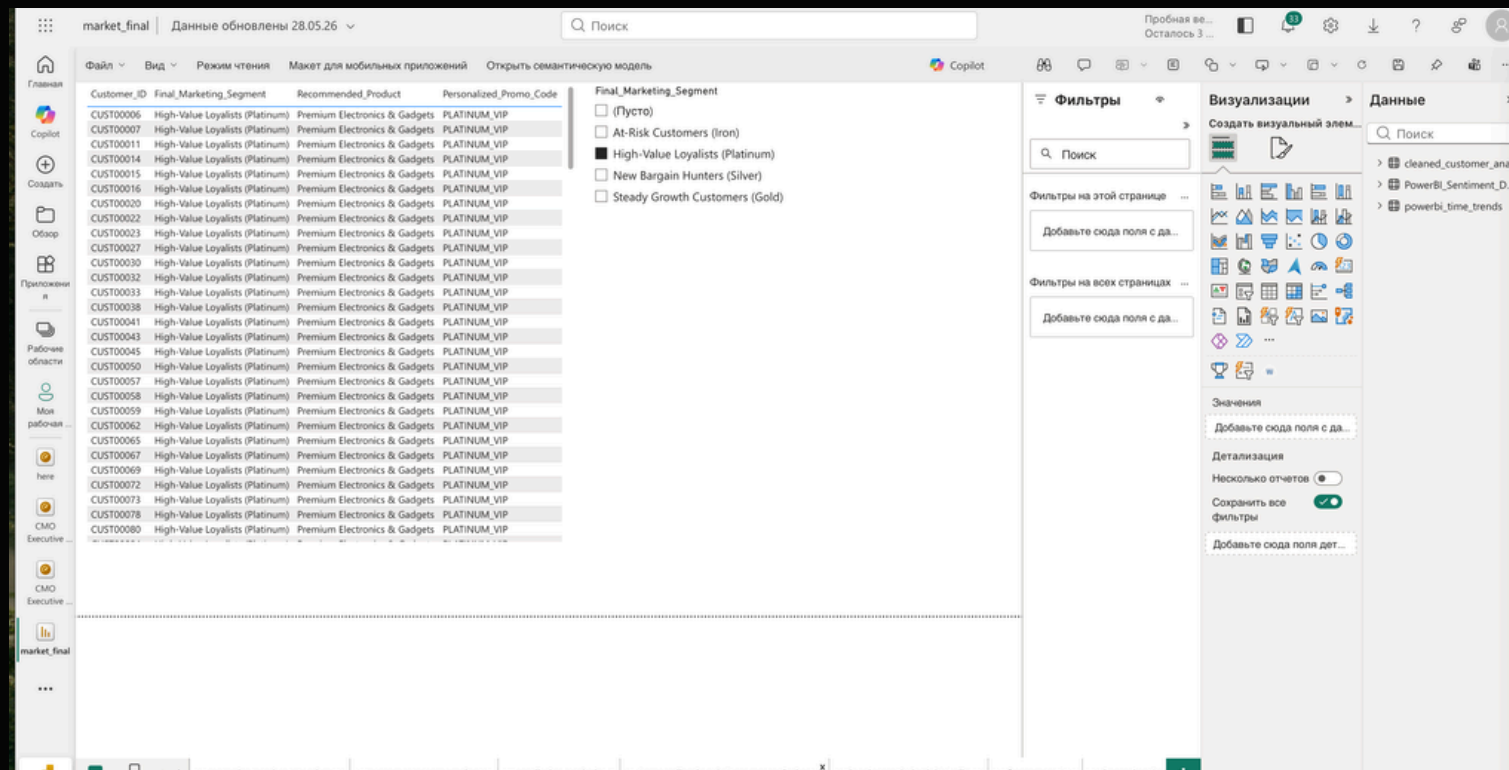


PREDICTIVE ANALYTICS DASHBOARD

""Page 3 showcases our predictive modeling outputs. Here, we transition from descriptive analysis to predictive intelligence using supervised machine learning. Our Churn Prediction Model analyzes historical customer signals to calculate a precise Churn_Probability score for every individual in the dataset. The Feature Importance Chart on this page explicitly reveals the underlying logic of our AI. It demonstrates to the marketing team that a decline in weekly session counts (Session_Count) paired with an increase in cart abandonment (Cart_Abandon_Rate) serves as the strongest operational trigger that a customer is about to leave the brand."



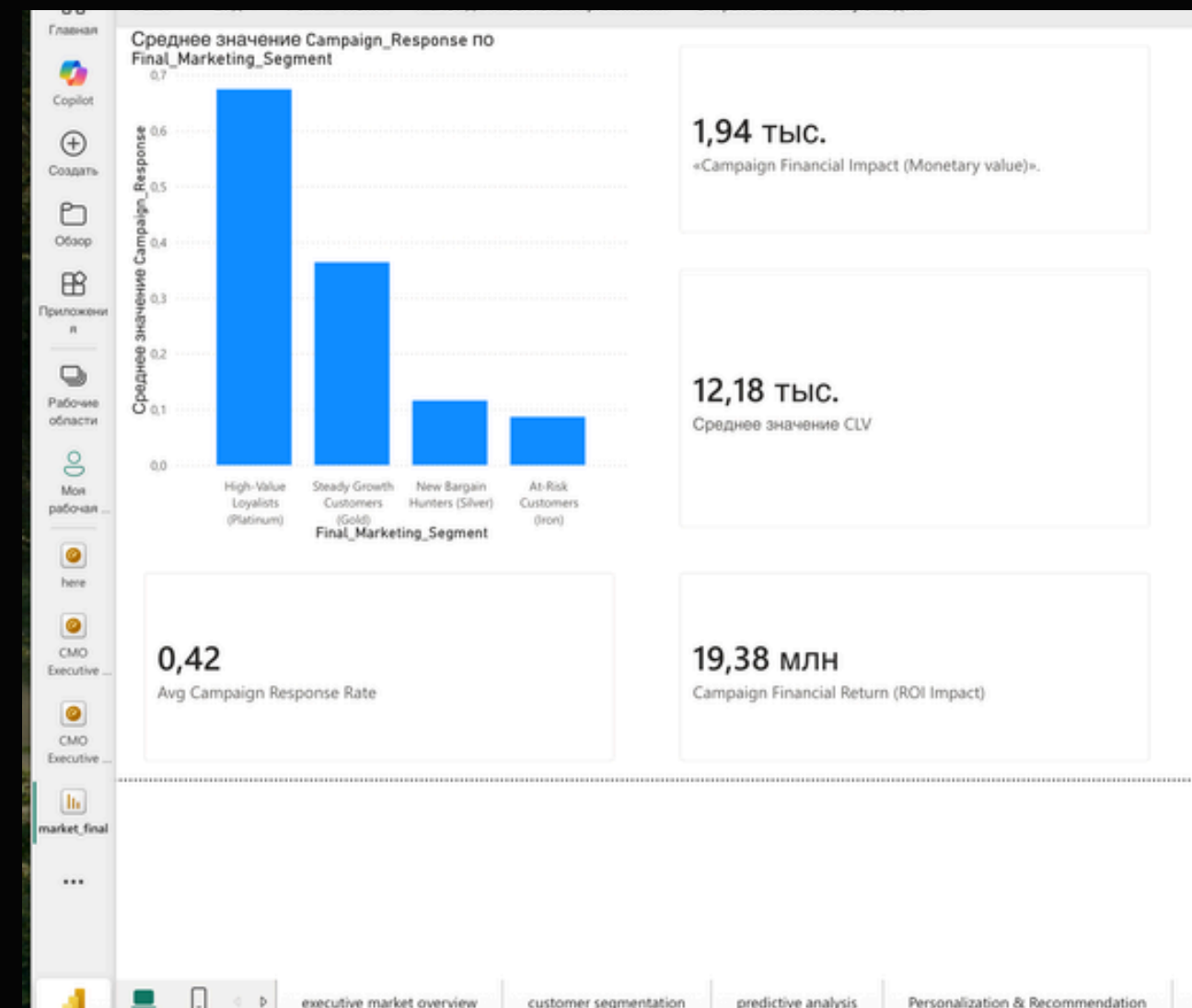
PERSONALIZATION & RECOMMENDATION DASHBOARD



"Page 4 demonstrates our AI Recommendation Engine at work. Utilizing collaborative filtering algorithms, the system maps purchasing habits within each K-Means cluster to generate a list of Top_Recommended_Products tailored to each segment. To justify this technologically, our side-by-side bar chart compares standard campaign performance against personalized campaign outcomes. The data clearly validates our approach: personalized recommendations driven by machine learning deliver a conversion rate that is significantly higher than generic mass marketing."

CAMPAIGN PERFORMANCE DASHBOARD

""On Page 5, we evaluate the financial effectiveness of our marketing initiatives. Since traditional media channel flags were not present in the raw behavioral tables, we took an advanced approach: we aggregated the Campaign_Response_Rate and financial returns across our AI-generated behavioral segments. Using DAX measures, we modeled our Campaign Financial Return (ROI Impact) based on total accumulated Monetary values. This allows the CMO to instantly pinpoint which customer groups yield the highest return on investment, shifting budget allocation from underperforming segments to high-yield clusters."



SENTIMENT & SOCIAL MEDIA DASHBOARD

"Page 6 brings our Natural Language Processing (NLP) pipeline to life. Using Python libraries such as spaCy and NLTK, we processed thousands of unstructured textual customer reviews, executing text cleaning, tokenization, lemmatization, and sentiment polarity scoring.

This visual distribution categorizes customer language into Positive, Neutral, and Negative sentiment. By tracking the average Polarity score alongside a customer word cloud, the dashboard isolates specific product issues and calculates a real-time Brand Health Score, making it immediately obvious where customers love the brand and where they face friction."



ETHICS, PRIVACY & RISK DASHBOARD

""A core pillar of our project is Responsible AI. First, our data privacy checklist confirms that all sensitive customer records and identifiers have been fully anonymized and hashed to ensure strict data protection compliance. Second, we feature an algorithmic Bias Detection Summary to guarantee model fairness. This ensures our churn and promotional models evaluate behavioral metrics purely on transaction data, preventing any discriminatory treatment or unfair discount distribution across demographic or income segments."

AI MARKETING STRATEGY & RECOMMENDATIONS

"To conclude, we present our final strategic roadmap to the CMO, backed by three undeniable insights from our data:

1. Focus on High-Value Clusters: Our K-Means model identified that while the 'Loyalist' segment represents a minority of the customer base, they generate the vast majority of our total Monetary value. We recommend reallocating 40% of the budget to VIP retention.
2. Mitigate Cart Abandonment: Customers showing an average Cart_Abandon_Rate of 17.65% are prime targets for churn. We recommend deploying automated, time-sensitive push notifications triggered by our recommendation engine.

